Ghost Gutzwiller for the Half-Filled Hubbard Dimer by Direct Minimization (No Impurity Problem): Equations and Implementation of `not_imp_gga_dimer.py`

gGut_mau project

October 19, 2026

Contents

1 The problem **2**

2 The variational functional **2**

3 Variables and parametrization **3**

4 Objective, constraints and analytic gradients **4**

5 Minimization **5**

6 Recovery of the multipliers and consistency checks **5**

7 Second-order test **6**

8 Results ($U/t = 2$, $N_g = 2$) and status **7**

9 Usage and correspondence with `gga_dimer.py` **8**

10 Reconstruction of R and λ from x and X **8**

11 Generalization: a site-splitting potential ΔV **10**

Purpose of this note. `gga_dimer.py` solves the ghost Gutzwiller (gGA) equations of the Hubbard dimer by the iterative embedding cycle of `code.tex`: an impurity Hamiltonian $H_{\rm imp}(V, \lambda^c)$ is diagonalized at every iteration. `not_imp_gga_dimer.py` is an *alternative* solver for *the same problem*: no impurity Hamiltonian, no V^I, no $\lambda^{I,c}$ and no fixed-point loop appear. Following Tagliente, Pasqua and Fabrizio (arXiv:2607.27156, `NotImpurity.pdf`, Eqs. 7 and 9–12), the variational energy functional is minimized *directly* over the impurity wavefunctions $|\phi^I\rangle$ and the quasiparticle Slater determinant, subject to the density constraints. The notation is that of `code.tex` ($N_{\rm eff} = 1 + N_g$, R^I, Δ^{II}, D^I, $\Delta^{I,\rm bb}$, ρ, λ^I, $S(\Delta)$, $\mathbf{I}$, X_{ab}, ...); equations of `code.tex` are cited by name, not by number.

1 The problem

Model. Exactly as in `code.tex`: the half-filled two-site Hubbard dimer with hopping $-t$ and local Hamiltonian $H_{\rm loc} = U\, n_\uparrow n_\downarrow - \mu\,(n_\uparrow + n_\downarrow)$, $\mu = U/2$ (it differs from the $\frac{U}{2}(n-1)^2$ of the reference paper by the constant $U/2$ per site, which is kept so that energies are directly comparable with `gga_dimer.py`). $N_{\rm phys} = 1$ physical orbital per fragment, $N_{\rm eff} = 1 + N_g$ quasiparticle (bath) orbitals per fragment and spin; N_g *is even*, so $N_{\rm eff}$ is odd.

Symmetry class. As in `gga_dimer.py`: paramagnetic (spin-restricted), the two fragments $I = 0, 1$ *not* assumed equal.

- The impurity wavefunction $|\phi^I\rangle$ is a *real* vector in the half-filled, spin-balanced sector $N_\uparrow = N_\downarrow = (1 + N_{\rm eff})/2$ of the embedding Fock space (dimension $4^{1+N_{\rm eff}}$), a sector of dimension

$$m \;=\; \binom{1+N_{\rm eff}}{(1+N_{\rm eff})/2}^2 \qquad (m = 36 \text{ for } N_g = 2). \tag{1}$$

  All operators entering below conserve $N_\uparrow$ and $N_\downarrow$ separately, so their restriction to this sector is exact (no leakage).

- The quasiparticle Slater determinant $|\psi_*\rangle$ has one 1-RDM $\rho \in \mathbb{R}^{2N_{\rm eff}\times 2N_{\rm eff}}$ (per spin, the same for both), a rank-$N_{\rm eff}$ projector: half filling, i.e. $N_{\rm eff}$ of the $2N_{\rm eff}$ levels are occupied (the filling rule of `code.tex`).

Blocks of ρ are written ρ^{IJ} (ρ^{II} is $\Delta^{II}_{\rm qp}$ of `code.tex`).

2 The variational functional

Energy (Eq. 11 of the reference). Specialized to the dimer and summed over spin and over the two bonds,

$$\boxed{E\big(\rho,\phi^0,\phi^1\big) \;=\; -4t\,\big(R^0\big)^T\rho^{01}R^1 \;+\; \sum_{I=0,1}\langle\phi^I|H_{\rm loc}|\phi^I\rangle.} \tag{2}$$

The factor 4 is 2 (spin) × 2 (the two ordered fragment pairs), the same factor as in the variational energy of `code.tex`. Here R^I is *not* an independent variable: it is the function $R(\phi^I)$ of Eqs. 9–10 of the reference,

$$R^I \;=\; S\big(\Delta^I\big)^{-1}D^I, \qquad \Delta^I \;\equiv\; \mathbf{I} - \Delta^{I,\rm bb}, \qquad S(\Delta) = \sqrt{\Delta(\mathbf{I}-\Delta) + \varepsilon\,\mathbf{I}}, \tag{3}$$

with the two expectation values over the impurity state (spin averages)

$$\Delta^{I,\rm bb}_{ab} = \tfrac{1}{2}\big(\langle d^\dagger_{a\uparrow}d_{b\uparrow}\rangle + \langle d^\dagger_{a\downarrow}d_{b\downarrow}\rangle\big)_{\phi^I}, \qquad D^I_a = \tfrac{1}{2}\big(\langle d^\dagger_{a\uparrow}c_\uparrow\rangle + \langle d^\dagger_{a\downarrow}c_\downarrow\rangle\big)_{\phi^I}. \tag{4}$$

These are the same $\Delta^{I,\rm bb}$ and D^I as in the impurity section of `code.tex`; $S(\Delta)$ is the shifted square root of `code.tex` ($\varepsilon = 10^{-9}$ by default, added to the eigenvalues of $\Delta(\mathbf{I}-\Delta)$). Since $\Delta \mapsto \mathbf{I} - \Delta$ leaves $\Delta(\mathbf{I}-\Delta)$ unchanged, $S(\mathbf{I}-\Delta^{I,\rm bb}) = S(\Delta^{I,\rm bb})$, i.e. Eq. (3) is the reference's $R = Q\,S^{-1}$ with $S^2 = \Delta_\phi(\mathbf{I}-\Delta_\phi)$, $\Delta_\phi = \Delta^{I,\rm bb}$, $Q = D$.

Constraints (Eq. 7 of the reference).

$$\langle\phi^I|\phi^I\rangle = 1, \qquad \rho^{II} \;=\; \mathbf{I} - \Delta^{I,\mathrm{bb}}\big(\phi^I\big) \qquad (I = 0, 1), \tag{5}$$

the second being the requirement that the local density matrix of the Slater determinant equal the particle–hole transform of the bath density matrix of the impurity state.

The functional F (Eq. 12) and the Lagrangian actually used. Eq. 12 of the reference is $F = E - E_*\langle\psi_*|\psi_*\rangle - E_\phi\langle\phi|\phi\rangle + \mathrm{Tr}\{\hat\lambda[\mathbf{I} - \hat\Delta_\phi - \hat\Delta_*]\}$: energy, two normalizations, and a single matrix multiplier $\hat\lambda$ for the density constraint. With our spin factors and eliminating the two normalizations (see below), the saddle point of F is the stationary point of

$$\mathcal{L} \;=\; E(\rho, \phi^0, \phi^1) \;+\; \sum_I 2\,\mathrm{Tr}\Big[\lambda^I\big(\mathbf{I} - \Delta^{I,\mathrm{bb}}(\phi^I) - \rho^{II}\big)\Big], \tag{6}$$

where λ^I is the same λ^I as in the quasiparticle Hamiltonian $H_{\mathrm{qp}} = \left[\begin{smallmatrix} -\lambda^0 & -tR^0R^{1T} \\ -tR^1R^{0T} & -\lambda^1 \end{smallmatrix}\right]$ of `code.tex`. Compared with the Lagrangian of `code.tex` (which also carries V^I, $\lambda^{I,c}$ and the independent variables R^I, Δ^I), the constraints associated with V^I ($D^I = S(\Delta^I)R^I$) and with $\lambda^{I,c}$ ($\Delta^I + \Delta^{I,\mathrm{bb}} = \mathbf{I}$) have been *substituted*: $R^I = R(\phi^I)$ and $\Delta^I = \mathbf{I} - \Delta^{I,\mathrm{bb}}$. This is the simplification of the direct approach.

3 Variables and parametrization

Unknowns. The unconstrained (redundant) variables are

$$z = \big(x^0, x^1, X\big), \qquad x^I \in \mathbb{R}^m, \quad X \in \mathbb{R}^{2N_{\mathrm{eff}} \times N_{\mathrm{eff}}}, \tag{7}$$

with the two parametrizations

$$\phi^I = \frac{x^I}{\|x^I\|}, \qquad \rho \;=\; X\,(X^TX)^{-1}X^T. \tag{8}$$

The first automatically enforces $\langle\phi^I|\phi^I\rangle = 1$ (so the norm multipliers E_ϕ never appear); the second is the orthogonal projector on the column space of X, a rank-N_{eff} projector for any full-rank X (so $|\psi_*\rangle$ is automatically a half-filled Slater determinant and E_* never appears). The redundancy is a gauge: $x^I \to c\,x^I$ and $X \to XA$ ($A \in GL(N_{\mathrm{eff}})$) leave everything invariant. The number of variables is $n_z = 2m + 2N_{\mathrm{eff}}^2$ ($= 90$ for $N_g = 2$).

Operator matrices (precomputed once). From the sparse Jordan–Wigner operators of `gga_dimer.py` (`build_impurity_ops`, mode order $c_\uparrow, c_\downarrow, d_{0\uparrow}, d_{0\downarrow}, \dots$) the following $m \times m$ real matrices are built by restriction to the sector:

$$\begin{aligned} H^s_{\mathrm{loc}} &= \mathrm{sector}(H_{\mathrm{loc}}), \\ \mathcal{Y}_a &= \mathrm{sector}\big(d^\dagger_{a\uparrow}c_\uparrow + c^\dagger_\uparrow d_{a\uparrow} + d^\dagger_{a\downarrow}c_\downarrow + c^\dagger_\downarrow d_{a\downarrow}\big), \\ B_{ab} &= \tfrac{1}{2}\,\mathrm{sector}\big(d^\dagger_{a\uparrow}d_{b\uparrow} + d^\dagger_{a\downarrow}d_{b\downarrow}\big) \;\text{ symmetrized in } (a,b), \end{aligned} \tag{9}$$

together with $n_\uparrow n_\downarrow$ and $n_\uparrow + n_\downarrow$ for the observables. For a real ϕ (so $\langle d^\dagger c\rangle = \langle c^\dagger d\rangle$) the expectation values of Eqs. (4) become plain quadratic forms:

$$\Delta^{I,\mathrm{bb}}_{ab} = \phi^T B_{ab}\,\phi, \qquad D^I_a = \tfrac{1}{4}\,\phi^T\mathcal{Y}_a\,\phi, \qquad E^I_{\mathrm{loc}} = \phi^T H^s_{\mathrm{loc}}\,\phi. \tag{10}$$

(The $\frac{1}{4}$ is $\frac{1}{2}$ from the spin average times $\frac{1}{2}$ from $\mathcal{Y}_a$ containing both $d^\dagger c$ and its transpose, for each spin.) This is the same evaluation as in `impurity_solve` of `gga_dimer.py`, except that the state ϕ is now a *variable* instead of the ground state of H_{imp}.

4 Objective, constraints and analytic gradients

Everything is differentiated analytically (checked against central finite differences by `--check-gradients`: agreement $\sim 2\times10^{-10}$ for both E and the constraints).

Evaluation of E (`energy`). For each fragment: $\phi^I, \Delta^{I,\mathrm{bb}}, D^I$ from Eqs. (8), (10); diagonalize $\Delta^I = \mathbf{I} - \Delta^{I,\mathrm{bb}} = W\,\mathrm{diag}(d_i)W^T$; $m_i = d_i(1-d_i)+\varepsilon$, $f_i = m_i^{-1/2}$; $R^I = W\big[f \circ (W^T D^I)\big]$; then Eq. (2). (A valid 1-RDM has all $d_i \in [0,1]$, so up to rounding $m_i > 0$; the shift ε covers rounding of d_i to slightly negative values.)

Gradient with respect to ρ and X.

$$dE = \mathrm{Tr}[G_\rho\, d\rho] + \ldots, \qquad G_\rho = \begin{pmatrix} 0 & -2t\,R^0 R^{1T} \\ -2t\,R^1 R^{0T} & 0 \end{pmatrix}, \tag{11}$$

and, from $d\rho = P_\perp\, dX\, Y X^T + XY\, dX^T P_\perp$ with $Y = (X^TX)^{-1}$, $P_\perp = \mathbf{I} - \rho$,

$$\frac{\partial E}{\partial X} = 2\,P_\perp\, G_\rho\, X\, Y. \tag{12}$$

Gradient with respect to R^I and through $R(\phi^I)$. The partial derivatives with respect to the (dependent) vectors R^I are

$$g_{R^0} = -4t\,\rho^{01} R^1, \qquad g_{R^1} = -4t\,(\rho^{01})^T R^0. \tag{13}$$

Write $R = f(M)D$ with $M = \Delta(\mathbf{I}-\Delta)+\varepsilon\mathbf{I}$ and $f(m) = m^{-1/2}$. In the eigenbasis of Δ, $\tilde{u} = W^T g_R$, $\tilde{v} = W^T D$, the Daletskii–Krein derivative of $f(M)$ is the divided-difference matrix

$$\mathcal{F}_{ij} = \frac{f(m_i)-f(m_j)}{m_i - m_j} = -\frac{1}{s_i s_j (s_i+s_j)}, \qquad s_i = \sqrt{m_i}, \tag{14}$$

which is written in the cancellation-free form on the right (valid also for $m_i = m_j$). Since $dM_{ij} = (1-d_i-d_j)\,d\Delta_{ij}$ in this basis and $d\Delta = -d\Delta^{I,\mathrm{bb}}$,

$$\begin{aligned} g_D &= f(M)\,g_R = W\,(f\circ\tilde{u}), \\ \Gamma &= \mathrm{sym}\big[(\tilde{u}\tilde{v}^T)\circ\mathcal{F}\big], \qquad g_{\mathrm{bb}} = -\,W\big[\Gamma\circ(1-d_i-d_j)\big]W^T. \end{aligned} \tag{15}$$

(This is the same structure as the function grad_S of `code.tex`, here applied as an adjoint.) Collecting the three routes by which ϕ^I enters (through $\Delta^{I,\mathrm{bb}}$, through D^I, and through E_{loc}),

$$g_\phi = 2\sum_{ab}(g_{\mathrm{bb}})_{ab}\,B_{ab}\phi + \tfrac{1}{2}\sum_a (g_D)_a\,\mathcal{Y}_a\phi + 2\,H^s_{\mathrm{loc}}\phi, \qquad \frac{\partial E}{\partial x^I} = \frac{1}{\|x^I\|}\big(g_\phi - (\phi\cdot g_\phi)\phi\big), \tag{16}$$

the last step being the projection through $\phi = x/\|x\|$.

Constraints and Jacobian (`constraints`, `constraint_jac`). The independent entries ($a \le b$) of the two symmetric matrix equations of Eq. (5),

$$c^I_{ab}(z) = \big(\rho^{II} + \Delta^{I,\mathrm{bb}} - \mathbf{I}\big)_{ab} = 0, \qquad a \le b,\ I = 0, 1 \tag{17}$$

give $n_c = N_{\mathrm{eff}}(N_{\mathrm{eff}} + 1)$ equations (= 12 for $N_g = 2$). Their gradients are

$$\frac{\partial c^I_{ab}}{\partial x^I} = \frac{1}{\|x^I\|}\big(\mathbf{1} - \phi\phi^T\big)\, 2B_{ab}\phi, \qquad \frac{\partial c^I_{ab}}{\partial X} = 2\, P_\perp\, G^{(I,ab)}\, X\, Y, \tag{18}$$

where $G^{(I,ab)}$ is the symmetric matrix that has $\frac{1}{2}$ at positions $(a,b),(b,a)$ (or 1 at (a,a) if $a = b$) inside the I-th diagonal block and zero elsewhere. (For $N_g = 2$ the constraint Jacobian has full rank 12 at the solutions examined, backward and forward.)

5 Minimization

Algorithm. The constrained problem

$$\min_z\ E(z) \quad \text{subject to} \quad c(z) = 0 \tag{19}$$

is solved by sequential quadratic programming (`scipy.optimize.minimize`, `method="SLSQP"`, `ftol`= 10^{-13}, up to 1000 iterations), passing the analytic gradient of E and the analytic Jacobian of c. A run is accepted as *feasible* if $\max|c| < 10^{-8}$; among several starts the lowest-energy feasible result is kept. No impurity Hamiltonian is ever diagonalized in the minimization.

Starting points.

1. *Random start* (`random_start`): x^I with independent Gaussian components; $X =$ the occupied orbitals of H_{qp} built from the default cold-start $R^0 = R^1 = (0.9, 0.05, 0.065, \dots)$, $\lambda = 0$ of `gga_dimer.py`, plus $0.05\times$ Gaussian noise.

2. *From an* (R, λ) *guess* (`start_from_R_lambda`; files `R.in`, `L.in` in the format of `gga_dimer.py`): $X =$ the N_{eff} lowest eigenvectors of $H_{\mathrm{qp}}(R,\lambda)$; $\Delta^{II} = \rho^{II}$, V^I from the Eq. 7 of `code.tex` and $\lambda^{I,c}$ from its Eq. 8; then the lowest 16 eigenstates of the sector matrix $H^s_{\mathrm{loc}} + \sum_a V^I_a \mathcal{Y}_a + \sum_{ab} \lambda^{I,c}_{ab}\, 2B_{ab}$ are examined and the one whose bath density best satisfies $\Delta^{I,\mathrm{bb}} = \mathbf{I} - \Delta^{II}$ is taken as x^I. (This single diagonalization is used *only* to construct the starting ϕ^I; choosing among the low-lying eigenstates matters for the "forward" seed, whose impurity ground state has swapped ghost occupations and would otherwise violate the constraints by ≈ 0.97, after which SLSQP restores feasibility far away, at $E \approx -0.80$.)

6 Recovery of the multipliers and consistency checks

At a solution the multipliers of Eq. (6) are recovered, and the point is checked against the equations of the embedding approach of `code.tex` (`analyse`).

Stationarity conditions. $\partial\mathcal{L}/\partial\rho = 0$ on the manifold of projectors and $\partial\mathcal{L}/\partial\phi^I = 0$ on the unit sphere give, with $K = \left[\begin{smallmatrix} 0 & -tR^0R^{1T} \\ -tR^1R^{0T} & 0 \end{smallmatrix}\right]$ the hopping part of $H_{\rm qp}$ and $\Lambda = \mathrm{diag}(\lambda^0, \lambda^1)$,

$$[\,K - \Lambda,\ \rho\,] = 0 \qquad \text{(i.e. } H_{\rm qp} = K - \Lambda,\ \rho \text{ an eigenprojector)}, \tag{20}$$

$$\nabla_\phi E - 4\sum_{ab} \lambda^I_{ab}\, B_{ab}\phi^I = \mu_I\, \phi^I \qquad (I = 0, 1). \tag{21}$$

Both are *linear* in the $2\,N_{\rm eff}(N_{\rm eff}+1)/2$ independent entries of λ^0, λ^1. They are solved *jointly* by linear least squares. (Using Eq. (21) alone is not sufficient: when a ghost is inert the vectors $B_{ab}\phi$ are linearly dependent, its minimum-norm solution is not unique and need not make ρ an eigenprojector of $H_{\rm qp}$; with the joint system the rank is 12/12 for $N_g = 2$.) The residuals of the joint system are reported as "KKT: ϕ stationarity" and "KKT: $[H_{\rm qp}, \rho]$".

Quasiparticle energies. With the recovered λ^I, $H_{\rm qp}(R, \lambda)$ is built and diagonalized; its $2N_{\rm eff}$ eigenvalues are the quasiparticle energies `eqp` of the scans of `code.tex`.

Check against the embedding equations. With $R^I = R(\phi^I)$ and the recovered λ^I, V^I and $\lambda^{I,c}$ are computed from Eqs. 7 and 8 of `code.tex`, and the residual

$$\left\| \left(H^s_{\rm imp} - \langle\phi^I|H^s_{\rm imp}|\phi^I\rangle\right)\phi^I \right\|_\infty, \qquad H^s_{\rm imp} = H^s_{\rm loc} + \sum_a V^I_a\, \mathcal{Y}_a + \sum_{ab} \lambda^{I,c}_{ab}\, 2B_{ab}, \tag{22}$$

is reported ("$H_{\rm imp}\phi = E\phi$"). A small value shows that ϕ^I is an eigenstate of the embedding Hamiltonian of `code.tex` built from the solution, i.e. that the solution of the direct minimization also solves the embedding equations. The observables $Z^I = R^{I\,T}R^I$, double occupancy and $n_{\rm phys}$ are evaluated from Eqs. (10).

7 Second-order test

For a stationary point with full-rank constraints, a strict local minimum of the constrained energy requires the Hessian of the Lagrangian to be positive definite on the tangent space of the constraints. `hessian_analysis` computes it as follows.

1. Multipliers μ of the constraints from $\nabla E + J^T\mu = 0$ (least squares), $J = \partial c/\partial z$; residual reported.

2. Hessian H of $E + \mu\cdot c$ with respect to all n_z variables by central differences (step $h = 10^{-6}$) of the analytic gradient $\nabla E + J(z)^T\mu$, symmetrized.

3. Tangent space = null space of J (dimension $n_z - n_c = 78$ for $N_g = 2$); eigenvalues of the reduced Hessian $Z_t^T H Z_t$.

The redundancies of the parametrization give exact zero modes: 2 (norms of x^I) $+N_{\rm eff}^2$ ($X \to XA$) $+N_{\rm eff}(N_{\rm eff}-1)$ (rotations of the ghost basis of the two fragments), i.e. 17 for $N_g = 2$. A strict minimum has all other 61 eigenvalues positive.

8 Results ($U/t = 2$, $N_g = 2$) and status

The two $U/t = 2$ solutions discussed in `code.tex` ("backward" and "forward", seeds in `examples/`) are used as tests.

	backward	forward
$E_{\rm var}/t$ (this code)	-3.1256614043	-3.1249999925
$E_{\rm var}/t$ (`gga_dimer.py`)	-3.1256614043	-3.1250000
$Z^0 = Z^1$	0.99991	0.9375
$\max\|c\|$	6×10^{-15}	8×10^{-15}
KKT: ϕ / $[H_{\rm qp}, \rho]$ / $H_{\rm imp}\phi = E\phi$	$3{\times}10^{-7}$ / $8{\times}10^{-9}$ / $1{\times}10^{-7}$	$6{\times}10^{-7}$ / $6{\times}10^{-8}$ / $3{\times}10^{-7}$
Hessian: negative / zero / positive	0 / 17 / 61	0 / 20 / 58
smallest positive eigenvalue	2.90	0.0060

Table 1: Solutions of the direct minimization at $U/t = 2$, $N_g = 2$. Backward: started from the (R, λ) seed of `examples/U2_backward`. Forward: converged from a random start (the seed of `examples/U2_forward` leads to the backward solution, see below). The energies agree with `gga_dimer.py`.

- **Backward** is reproduced to 10^{-12} and is a strict local minimum (the 61 positive eigenvalues are all the non-gauge ones; the softest is 2.90).
- **Forward**: unlike the fixed point of `gga_dimer.py` (where ρ^{II} and $\mathbf{I} - \Delta^{I,\rm bb}$ differ by ≈ 0.97 with swapped ghost occupations), here ρ^{II} and $\mathbf{I} - \Delta^{I,\rm bb}$ coincide entry by entry (10^{-12}), both with eigenvalues $\{0, \frac{1}{2}, 1\}$: the constraints of Eq. (5) hold. Multipliers exist (KKT residuals $\lesssim 10^{-6}$). The reduced Hessian has no negative eigenvalue but 3 zero modes in addition to the 17 gauge ones, and its positive eigenvalues are small (0.0060 smallest). Along all 20 zero-mode directions, followed on feasible curves, E never decreases and grows as s^4 (about 3×10^{-6} at $s = 0.2$); thus the second-order test is *inconclusive along those* 3 *directions*. The origin of these extra flat directions was not investigated. An independent empirical test (random displacements projected back on the constraints, 4000 of them, steps 10^{-3}–0.1) finds only energy increases growing as s^2 with coefficient $\approx 7 \times 10^{-3}$.
- **The forward seed does not stay on the forward minimum.** A single SLSQP pass started from the forward seed stops early ($E = -3.12499887$, projected gradient 4×10^{-3}, descent directions with $\Delta E \simeq -8 \times 10^{-5}$ exist). A second pass continues downhill, *not* to the forward minimum -3.1249999925 but to the lower ghost-assisted (backward) solution -3.1256614043, and further passes stay there. The forward minimum listed in Table 1 was reached from a random start; it is a local minimum with a small basin, and the seed lies outside it. Consequently the "forward" U-scan of `scan_not_imp_U.py`, seeded from that point, converges to the ghost-assisted branch at $U/t = 2$ (it stays on the plain-Gutzwiller branch only for $U/t \lesssim 0.8$).

Limitation: random starts. From random starts the minimization always reaches the *higher* boundary minima with inert ghosts, never the ghost-assisted solutions: 120/120 starts give $E = -3.125$ at $U/t = 2$ (versus -3.125661), 30/30 give -10.0 at $U/t = 10$ (versus -10.101314 from `gga_dimer.py`), 30/30 give -6.0 at $U/t = 6$; at $U/t = 4$ the minima are -4.5 and -4.0. Perturbing the boundary minimum by up to 0.3 in the variables and re-minimizing also returns to it. The ghost-assisted solutions are reached only from a seed close to them (as with the backward seed

above). Making the direct minimization find them without a seed (for example by continuation in U, warm-starting (x^I, X) from the previous U) has *not* been done.

9 Usage and correspondence with `gga_dimer.py`

```
python3 not_imp_gga_dimer.py --U <U/t> --Ng <even> [--t 1.0] [--nstarts 10] [--seed 0]
                             [--R-file R.in --lam-file L.in] # seeded start (else random)
                             [--max-iter 1000] [--shift 1e-9] [--quiet]
                             [--omega <w_max> --eta <eta> --domega <step>] # spectral
                                function
python3 not_imp_gga_dimer.py --check-gradients [--Ng 2 --U 2] # analytic vs finite diff.
```

It prints Z^I, E_{var}/t, the double occupancy, the constraint violation, the three KKT residuals and `eqp`, and writes `not_imp_Ng*_U*_{R,lambda,lambda_c}.dat` in the layout of `gga_dimer.py`'s `save_results`.

Spectral function. If `--omega`, `--eta` and `--domega` are all given (any subset is an error), the spectral function of the solution is computed exactly as in the post-processing section of `code.tex`: $G(\omega) = [(\omega + i\eta)\mathbf{I} - H_{\mathrm{qp}}]^{-1}$ with the H_{qp} of the reconstructed (R, λ) (Section 10), then $G_{\mathrm{phys}} = R_{\mathrm{mat}}^T G\, R_{\mathrm{mat}}$ with $R_{\mathrm{mat}} = \mathrm{diag}(R^0, R^1)$ and $A(\omega) = -\frac{2}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,G_{\mathrm{phys}}(\omega)$ on the grid $-\omega_{\max}, \ldots, \omega_{\max}$ in steps of $d\omega$, written to `A_omega.txt`; the routine is the one of `gga_dimer.py` (`compute_spectral_function`), and the sum rule $\int A\, d\omega \to 4$ is printed. The only input from the minimization is (R, λ), so $A(\omega)$ inherits the ambiguity of the nearly free direction of λ described in Section 10; at $U/t = 10$ (the `examples/U10` seed, $\omega_{\max} = 15$, $\eta = d\omega = 0.05$) it gives $\int A\, d\omega = 3.9615$ and agrees with the $A(\omega)$ of `gga_dimer.py` for the same case to 6×10^{-5} (maximum of A: 6.48).

	`gga_dimer.py` (embedding)	`not_imp_gga_dimer.py` (direct)
unknowns	R^I, λ^I (and $V^I, \lambda^{I,c}, \Delta^{II}$ derived)	x^I (ϕ^I), X (ρ)
R^I	output of the impurity solve, $S(\Delta^{\mathrm{new}})^{-1} D$	function of ϕ^I, Eq. (3)
V^I, $\lambda^{I,c}$	Eqs. 7, 8 of `code.tex`, needed in every iteration	not used; only in the a-posteriori check
impurity ground state	diagonalized every iteration	never (only for a seeded start)
constraint $\rho^{II} = \mathbf{I} - \Delta^{I,\mathrm{bb}}$	*not enforced* by the convergence test	enforced ($\max \lvert c \rvert \sim 10^{-14}$)
solver	damped fixed-point iteration	SLSQP on E with equality constraints
result type	fixed point of the iteration	constrained local minimum (checked)

Table 2: Correspondence between the two solvers.

10 Reconstruction of R and λ from x and X

The unknowns of the direct minimization are only $z = (x^0, x^1, X)$; neither R^I nor λ^I is a variable. Both are reconstructed from a solution in the function `analyse`, in the following order. Let $n_s = N_{\mathrm{eff}}(N_{\mathrm{eff}} + 1)/2$ (the number of independent entries of a symmetric $N_{\mathrm{eff}} \times N_{\mathrm{eff}}$ matrix; $n_s = 6$ for $N_g = 2$).

Step 1: the states.

$$\phi^I = \frac{x^I}{\|x^I\|}, \qquad \rho = X\,(X^TX)^{-1}X^T, \qquad \rho^{IJ} = \rho_{[IN_{\rm eff}:(I+1)N_{\rm eff},\, JN_{\rm eff}:(J+1)N_{\rm eff}]}. \tag{23}$$

Step 2: R^I (explicit). R^I is a function of ϕ^I alone, Eqs. (3) and (10):

$$\Delta^{I,\rm bb}_{ab} = \phi^{I\,T} B_{ab} \phi^I, \qquad D^I_a = \tfrac{1}{4}\,\phi^{I\,T}\mathcal{Y}_a\phi^I, \qquad \mathbf{I} - \Delta^{I,\rm bb} = W\,\mathrm{diag}(d_i)\,W^T,$$
$$R^I = W\Big[\big(d_i(1-d_i)+\varepsilon\big)^{-1/2} \circ \big(W^T D^I\big)\Big]. \tag{24}$$

The matrix ρ plays no role here. $Z^I = R^{I\,T}R^I$.

Step 3: λ^I (by linear least squares). The multiplier λ^I is not determined by (x, X) through a formula but by the two stationarity conditions of Eq. (6), Eqs. (20) and (21), which are *linear* in the entries of λ^0, λ^1. Collect them in the unknown vector

$$w = \big(\lambda^0_{ab},\ \lambda^1_{ab}\big)_{a\le b} \in \mathbb{R}^{2n_s}. \tag{25}$$

Let $\kappa_{ab} = 1$ for $a = b$ and $\kappa_{ab} = 2$ for $a < b$ (because $\sum_{ab}\lambda_{ab}B_{ab} = \sum_{a\le b}\kappa_{ab}\lambda_{ab}B_{ab}$), $P_I = \mathbf{1} - \phi^I\phi^{I\,T}$, $K = \left[\begin{smallmatrix} 0 & -tR^0R^{1T} \\ -tR^1R^{0T} & 0 \end{smallmatrix}\right]$ (with the R^I of Step 2), and let $\Xi^{(I,ab)}$ be the $2N_{\rm eff}\times 2N_{\rm eff}$ symmetric matrix with 1 at positions $(IN_{\rm eff}+a,\, IN_{\rm eff}+b)$ and $(IN_{\rm eff}+b,\, IN_{\rm eff}+a)$ (a single 1 if $a = b$) and zero elsewhere. The linear system is

$$\begin{pmatrix} A^{(0)} & 0 \\ 0 & A^{(1)} \\ C^{(0)} & C^{(1)} \end{pmatrix} w \;=\; \begin{pmatrix} b^{(0)} \\ b^{(1)} \\ \mathrm{vec}\,[K,\rho] \end{pmatrix}, \tag{26}$$

a system of $2m + 4N_{\rm eff}^2$ equations for the $2n_s$ unknowns, with

$$A^{(I)}_{\cdot,(ab)} = 4\,\kappa_{ab}\,P_I\big(B_{ab}\,\phi^I\big) \in \mathbb{R}^m, \qquad b^{(I)} = P_I\,g^{(I)}_\phi,$$
$$C^{(I)}_{\cdot,(ab)} = \mathrm{vec}\big[\Xi^{(I,ab)},\rho\big] \in \mathbb{R}^{4N_{\rm eff}^2}, \tag{27}$$

where $g^{(I)}_\phi$ is the gradient of E with respect to ϕ^I at fixed ρ (Eq. (16), including the dependence through $R(\phi^I)$), and the projector P_I removes the norm multiplier μ_I of Eq. (21). The first two block rows are the ϕ^I conditions, the last block row is $[\Xi,\rho]$-form of Eq. (20) ($[K-\Lambda,\rho]=0 \Leftrightarrow [\Lambda,\rho] = [K,\rho]$). The system is solved in the least-squares sense (`numpy.linalg.lstsq`, `rcond`$= 10^{-6}$, minimum-norm solution), and λ^I is the symmetric matrix built from the n_s entries of w belonging to fragment I.

- *Why jointly.* The ϕ block alone is rank deficient whenever a ghost is inert (then the vectors $P_I B_{ab}\phi^I$ are linearly dependent), so its minimum-norm solution is not unique and need not satisfy $[\,K-\Lambda,\rho\,]=0$; this happened for the forward solution ($[H_{\rm qp},\rho]\approx 0.16$). With the C block included the system has rank $2n_s = 12$ for $N_g = 2$ at the backward and forward solutions examined.

- *Nearly free direction.* For $U/t \gtrsim 7$ ($N_g = 2$) the smallest singular value of the system is $\sim 10^{-7}$, while the next one is ~ 0.15: one direction of λ is almost undetermined by the stationarity conditions (it belongs to the nearly decoupled bath orbital, whose quasiparticle

level is barely constrained). With a cutoff as small as 10^{-10} that direction is fixed by rounding noise, the residual is unchanged, and the two middle quasiparticle levels jump erratically with U (values between -3 and $+4$ were seen). The cutoff `rcond` $= 10^{-6}$ (relative to the largest singular value) removes it, i.e. takes the minimum-norm λ; results are identical for 10^{-6} and 10^{-4}, and the two levels then vary smoothly and stay at $\approx 10^{-3}$ (at the Fermi level, as expected for a decoupled bath) with the residual $\lesssim 8 \times 10^{-7}$. The position of those two levels is therefore a *choice* of the minimum-norm solution rather than something determined by (x, X) alone.

- *Quality check.* If (x, X) is a stationary point, the residual of Eq. (26) vanishes: it is reported as "KKT: ϕ stationarity" (the A rows) and "KKT: $[H_{\rm qp}, \rho]$" (the C rows, recomputed with the final $H_{\rm qp}$). Values found: $\lesssim 6 \times 10^{-7}$ and $\lesssim 6 \times 10^{-8}$ (Table 1). A large residual means that (x, X) is *not* a stationary point of $\mathcal{L}$, and the λ obtained is then only the best least-squares one.

Step 4: what follows. $H_{\rm qp} = K - \mathrm{diag}(\lambda^0, \lambda^1)$ is built as in `code.tex`, giving the quasiparticle energies `eqp`; V^I and $\lambda^{I,c}$ follow from Eqs. 7 and 8 of `code.tex` with ρ, R^I, $\Delta^I = \mathbf{I} - \Delta^{I,\mathrm{bb}}$ and λ^I. The arrays R^I, λ^I, $\lambda^{I,c}$ are written by `save_results` in the layout used by `gga_dimer.py` (two rows of $N_{\rm eff}$ numbers for R; two stacked $N_{\rm eff} \times N_{\rm eff}$ blocks for each matrix), so the files have the format of `R.in`/`L.in`.

Gauge. R^I and λ^I are obtained in the basis defined by x^I (the ghost-orbital basis of the impurity state) and X; no gauge fixing is applied. Solutions related by a rotation $Q_I \in O(N_{\rm eff})$ of the ghost basis of fragment I have $R^I \to Q_I R^I$ and $\lambda^I \to Q_I \lambda^I Q_I^T$, while E, Z^I and the quasiparticle energies are invariant. (The inverse map, from (R, λ) to a starting (x, X), is the seeded start described in Section 5.)

11 Generalization: a site-splitting potential ΔV

As in `gga_dimer.py` (`doc/code.tex`, Section "Generalization: a site-splitting potential ΔV"), the physical Hamiltonian can be generalized to $H_{\rm phys} \to H_{\rm phys} + \frac{\Delta V}{2}(n_0 - n_1)$ ($\Delta V = 0$ recovers the plain dimer; CLI flag `--deltaV` on `not_imp_gga_dimer.py` too). This is a bare one-body, fragment-diagonal term ($\epsilon_0 = +\Delta V/2$, $\epsilon_1 = -\Delta V/2$), so by the same Lagrangian argument as `code.tex`'s ΔV section, it adds $\sum_I \epsilon_I (R^I)^T \rho^{II} R^I$ to the energy functional of Eq. 11 and does *not* touch $H_{\rm loc}$ or the constraints (Eq. 7).

Energy and gradient (Section 2). Since this term is, like the kinetic term, linear in ρ (via its fragment-diagonal blocks ρ^{00}, ρ^{11} instead of the off-diagonal ρ^{01}), it is folded into the *same* matrix G of Eq. (12) (now also given diagonal blocks $G^{00} = \epsilon_0 R^0 (R^0)^T$, $G^{11} = \epsilon_1 R^1 (R^1)^T$), so the existing projector-derivative formula for $\nabla_X E$ needs no change; only $\nabla_{R^I} E$ (hence, through the chain rule of Section 4, $\nabla_{\phi^I} E$) gains the explicit extra term $2\epsilon_I \rho^{II} R^I$. Checked against finite differences (`check_gradients`, now with an optional `deltaV`): $\max|\text{analytic} - \text{FD}| \lesssim 10^{-10}$, same order as at $\Delta V = 0$.

Reconstruction (Section 10). Step 2 (R^I from ϕ^I) is unchanged – ΔV does not enter $H_{\rm loc}$ or ϕ^I's own energy, only ρ and R^I together, which are not yet coupled at that step. Step 3 changes

only in K, which must now carry the ΔV part of $H_{\rm qp}$'s diagonal blocks too (exactly $H_{\rm qp}$'s diagonal blocks of `code.tex`'s ΔV section minus λ):

$$K \;=\; \begin{pmatrix} \frac{\Delta V}{2} R^0 (R^0)^T & -t R^0 R^{1T} \\ -t R^1 R^{0T} & -\frac{\Delta V}{2} R^1 (R^1)^T \end{pmatrix}, \tag{28}$$

so that $\mathrm{vec}[K, \rho]$ (the right-hand side of the last block row of Eq. (26)) is the full one-body commutator target; $A^{(I)}$, $C^{(I)}$ are unchanged (the λ-dependence of $[H_{\rm qp}, \rho]$ does not involve ΔV). Step 4 (V^I, $\lambda^{I,c}$, $H_{\rm qp}$) calls `gga_dimer.py`'s `hybridization_V`/`embedding_lambda_c`/`qp_state`, now passing `deltaV` through, so it automatically uses the generalized V-fit of `code.tex`'s ΔV section.

A bug found and fixed. The first version of this generalization updated K above but left `hybridization_V` itself unchanged (i.e. called it without `deltaV`, implicitly $\epsilon_I = 0$ in the generalized V-fit). This is exactly the same omission described in `code.tex` ("A bug found and fixed while adding this"), and this document's own `analyse` machinery is precisely what exposed it: with K fixed but V^I not, "KKT: ϕ stationarity" and "KKT: $[H_{\rm qp}, \rho]$" were both small ($\lesssim 10^{-7}$, confirming λ^I itself was correctly recovered), yet "KKT: $H_{\rm imp}\phi = E\phi$" (which uses V^I, $\lambda^{I,c}$ built downstream of λ^I) was ~ 0.3 – a residual two orders of magnitude above the $\Delta V = 0$ baseline, and unchanged whether the point was reached by a random start or by seeding `minimize` exactly at a converged `gga_dimer.py` fixed point. This three-way split (two KKT checks fine, the third not) is what localized the bug to the V^I reconstruction specifically, rather than to the energy functional or to λ^I's own recovery (both already independently validated by the finite-difference gradient check and by `hessian_analysis`'s simpler, direct Lagrange-stationarity test, $\|\nabla E + J^T \mu\|_\infty \sim 10^{-8}$, which does not depend on `hybridization_V` at all). After passing `deltaV` through, "KKT: $H_{\rm imp}\phi = E\phi$" dropped to $\sim 10^{-7}$, matching the other two residuals and the $\Delta V = 0$ baseline.

Validation summary. With the fix: finite-difference gradient check $\lesssim 10^{-10}$; `hessian_analysis` stationarity $\sim 10^{-8}$ with no negative reduced-Hessian eigenvalue (genuine local minimum, up to the documented zero modes); all three KKT residuals $\sim 10^{-7}$–10^{-8}, same order as $\Delta V = 0$; and, for $N_g = 0$ (several $(U, \Delta V)$), agreement with `gga_dimer.py` to $\sim 10^{-8}$–10^{-10} in $E_{\rm var}$, Z^0, Z^1. For $N_g = 2$, `gga_dimer.py`'s fixed-point iteration was found to converge to a spurious, much higher-energy "sleeping ghost" solution regardless of the starting guess (including when seeded exactly at this script's converged, KKT-validated solution), so the two solvers were *not* cross-checked at $N_g = 2, \Delta V \neq 0$ beyond confirming (via the KKT residuals above) that *this* script's own solution is correct; see `code.tex`'s "Known caveat" paragraph for how this relates to the pre-existing $\Delta V = 0$ multi-fixed-point phenomenon.

References

[1] A. M. Tagliente, I. Pasqua and M. Fabrizio, *Direct minimization versus iterative embedding in the ghost-Gutzwiller method: a comparative study of magnetism in Mott insulators*, arXiv:2607.27156 (`NotImpurity.pdf`); Eqs. 7 and 9–12 are the basis of this note.

[2] `doc/code.tex`: notation and equations of the embedding solver `gga_dimer.py`, used throughout.